Day 19 – Agile & Scrum

1. What is Agile?

Agile is a software development approach in which software is developed and delivered in small parts through continuous feedback and improvements.

Example – Banking Application: A banking application can be developed in small parts, such as Login, Balance Enquiry, Fund Transfer and Payment functionality.

2. What is Scrum?

Scrum is an Agile framework used to manage and develop software through short iterations called Sprints.

Sprint: A fixed-length period, usually 1 to 4 weeks, during which the Scrum Team works toward a Sprint Goal and creates a usable product increment.

3. Scrum Activities

A. Sprint Planning

Sprint Planning is a meeting where the Scrum Team decides why the Sprint is valuable, what work can be completed, and how the selected work will be done.

Example: The team plans Login and Registration functionality for the upcoming Sprint. Developers plan development work, and testers plan testing activities.

B. Daily Stand-up (Daily Scrum)

Daily Scrum is a short daily event where Developers inspect progress toward the Sprint Goal and adapt their plan. The team may discuss progress, today's plan and blockers.

Example: A tester says: Yesterday I completed login testing. Today I will test registration. I am waiting for test environment access.

C. Sprint Review

Sprint Review is held near the end of the Sprint to inspect the Sprint outcome with stakeholders and discuss feedback and future changes.

Example: The team demonstrates completed Login and Registration features to the Product Owner and stakeholders and collects feedback.

D. Sprint Retrospective

Sprint Retrospective is held after the Sprint Review and before the next Sprint Planning. The Scrum Team discusses how to improve quality and effectiveness.

Example: The team discusses what went well, what problems occurred and what improvements they will try in the next Sprint.

4. Difference Between Scrum Activities

Activity	Main Purpose
Sprint Planning	Plan the Sprint Goal and selected work.
Daily Stand-up	Inspect daily progress and adapt the plan.
Sprint Review	Inspect the product outcome and collect stakeholder feedback.
Sprint Retrospective	Identify ways to improve the team's work and quality.

5. Quick Revision

• **Agile:** A software development approach based on incremental delivery, feedback and adaptation.

• **Scrum:** An Agile framework.

• **Sprint:** A fixed-length period for achieving a Sprint Goal and creating a usable increment.

• **Sprint Planning:** Plan the Sprint.

• **Daily Scrum:** Inspect progress and adapt the daily plan.

• **Sprint Review:** Inspect the product outcome and gather feedback.

• **Sprint Retrospective:** Improve the team's way of working.